

CASEY STATION, Antarctica

WMO: 896110

Lat: 66.282S

Lon: 110.523E

Elev: 139

StdP: 14.62

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	-20.2	-16.8	-27.8	1.2	-19.0	-24.5	1.4	-16.2	88.5	16.4	83.1	14.8	13.9	170	1.214

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.2	38.1	32.8	36.7	32.0	35.3	31.2	33.5	36.8	32.6	35.6	31.6	34.5	15.8	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
30.7	25.0	33.1	29.7	23.9	32.7	28.6	22.7	32.4	12.5	37.2	12.1	35.9	11.7	34.7	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
75.0	65.7	55.9	-24.3	41.6	3.5	2.1	-26.9	43.2	-28.9	44.4	-30.9	45.6	-33.5	47.1		
			-24.6	35.2	3.5	1.4	-27.1	36.2	-29.2	37.0	-31.1	37.8	-33.7	38.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.9	31.1	26.8	18.9	11.2	5.8	5.8	5.4	5.0	7.2	10.7	22.0	29.5	(6)
(7)		DBStd	12.76	3.07	4.33	6.65	9.12	10.95	10.64	11.10	10.79	9.67	7.43	5.68	3.50	(7)
(8)		HDD50	12820	585	651	965	1163	1371	1327	1383	1394	1284	1220	841	636	(8)
(9)		HDD65	18295	1050	1071	1430	1613	1836	1777	1848	1859	1734	1685	1291	1101	(9)
(10)		CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(10)
(11)		CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	16.0	12.1	13.4	15.2	16.8	16.1	17.4	19.3	18.4	18.7	15.6	15.6	13.2	(14)
(15)	Precipitation	PrecAvg	18.0	1.5	1.6	1.6	1.4	1.3	1.5	1.8	1.8	1.4	1.2	1.5	1.5	(15)
(16)		PrecMax	23.1	2.2	2.6	2.8	2.8	2.1	2.7	3.1	3.0	2.4	2.0	2.8	2.8	(16)
(17)		PrecMin	14.3	0.8	0.2	0.7	0.5	0.3	0.5	0.8	0.7	0.6	0.4	0.4	0.6	(17)
(18)		PrecStd	2.2	0.4	0.6	0.5	0.6	0.4	0.5	0.7	0.6	0.4	0.4	0.6	0.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.5	38.0	33.4	31.2	30.4	27.9	30.0	30.3	29.6	28.5	35.3	40.1	(19)
(20)			MCWB	34.2	32.9	29.7	28.4	28.3	25.8	28.0	28.6	27.6	25.6	30.4	33.5	(20)
(21)		2%	DB	38.7	36.0	31.0	28.4	27.4	25.0	26.6	26.2	25.3	25.9	32.5	37.6	(21)
(22)			MCWB	33.1	31.7	27.8	26.0	25.7	23.2	25.1	24.6	23.2	23.4	28.7	32.3	(22)
(23)		5%	DB	37.3	34.3	29.4	25.8	24.3	22.8	23.4	23.3	22.7	23.8	31.0	36.0	(23)
(24)			MCWB	32.4	30.5	26.7	23.5	22.7	21.1	21.5	21.7	20.7	21.5	27.7	31.4	(24)
(25)		10%	DB	35.9	32.8	27.6	23.6	21.3	20.5	20.5	20.0	20.3	21.8	29.5	34.5	(25)
(26)			MCWB	31.6	29.4	25.3	21.6	19.7	18.9	18.9	18.4	18.6	19.7	26.5	30.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.0	34.0	30.4	29.0	29.0	26.7	28.8	29.0	28.2	26.2	31.4	34.5	(27)
(28)			MCDWB	39.0	36.8	32.4	30.4	30.4	27.6	29.3	30.3	29.1	27.6	33.6	38.5	(28)
(29)		2%	WB	33.9	32.4	28.4	26.3	25.7	23.4	25.3	24.9	23.5	23.6	29.5	33.0	(29)
(30)			MCDWB	37.3	35.1	30.6	28.3	27.1	24.7	26.4	26.2	25.0	25.5	31.8	36.3	(30)
(31)		5%	WB	33.0	31.0	26.8	23.8	22.7	21.2	21.5	21.9	20.9	21.7	28.1	32.0	(31)
(32)			MCDWB	36.2	33.7	29.0	25.5	24.2	22.6	23.2	23.2	22.5	23.6	30.6	35.0	(32)
(33)		10%	WB	32.1	29.7	25.4	21.7	19.8	19.0	18.9	18.5	18.6	19.7	26.7	31.0	(33)
(34)			MCDWB	35.2	32.4	27.5	23.4	21.2	20.4	20.5	19.9	20.3	21.6	29.2	33.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.2	6.7	7.9	9.5	10.1	10.3	10.8	10.2	9.7	9.4	8.4	6.6	(35)
(36)			MCDBR	7.7	7.6	7.7	9.5	11.1	10.1	10.8	10.6	9.3	8.6	8.7	7.8	(36)
(37)			MCWBR	5.0	5.5	6.2	8.6	10.1	9.2	9.8	9.7	8.3	7.4	6.9	5.2	(37)
(38)		5% WB	MCDBR	6.6	6.8	7.5	9.4	11.6	10.3	10.6	10.8	9.3	8.5	8.1	6.8	(38)
(39)			MCWBR	4.5	5.1	6.3	8.9	10.9	9.7	9.9	10.2	8.6	7.5	6.7	4.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.267	0.256	0.234	0.194	0.129	0.129	0.134	0.191	0.221	0.230	0.251	0.269	(40)	
(41)		taud	2.344	2.384	2.365	2.224	1.925	1.925	1.898	2.167	2.272	2.287	2.298	2.290	(41)	
(42)		Ebn at Noon	316	302	279	224	115	21	98	218	277	310	320	319	(42)	
(43)		Edh at Noon	34	29	22	14	4	1	4	14	23	31	36	37	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2443	1693	892	305	53	3	22	183	669	1500	2336	2739	(44)	
(45)		RadStd	108	93	48	23	3	0	1	12	30	44	83	93	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page